

IFYME004 2023/24

NCUK
THE UNIVERSITY CONSORTIUM

MATHEMATICS (ENGINEERING)

INTERNATIONAL FOUNDATION YEAR (IFY)



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SYLLABUS OVERVIEW

Module Code	IFYME004
Module Name	Mathematics Engineering
Programme Name	International Foundation Year
Percentage breakdown of Coursework	0%
Percentage breakdown of Exam/Test	100%
Delivery period	The syllabus will usually be delivered over two 15-week semesters
Recommended minimum teaching hours	120 (4 hours per week for 30 weeks)
Recommended minimum independent study hours	120 (4 hours per week for 30 weeks)

AIMS

To develop student's key knowledge, understanding, skills and application of mathematics appropriate for entry to a degree course at an NCUK University.

General Aims

- ⇒ To develop abilities to think logically, to recognise incorrect reasoning and to express ideas clearly.
- ⇒ To develop an enthusiasm for the subject and the skills required to apply the knowledge to both the further study and application of mathematics.
- ⇒ To develop in students an understanding of how theory and application work together.
- ⇒ To develop students' skills in modelling and the interpretation of results.
- ⇒ To develop the necessary English mathematics vocabulary and terminology to use their mathematics knowledge effectively in an NCUK University.
- ⇒ To encourage students towards a level of independence in both the planning and organisation of their studies.
- ⇒ To assist the development of competence and confidence of the students as learners, taking responsibility for their own learning through directed reading and study.

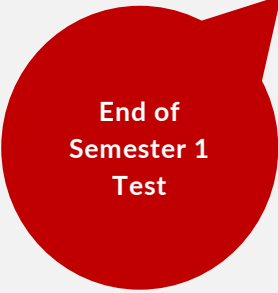
Specific Aims

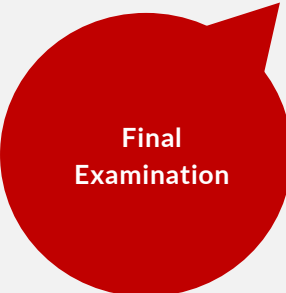
- ⇒ To revise basic skills and develop further skills in algebra.
- ⇒ To demonstrate basic skills in trigonometry and co-ordinate geometry.
- ⇒ To differentiate and integrate, including the selection and use of appropriate rules and techniques, and the application of calculus.
- ⇒ To solve simple separable differential equations, including the exponential decay and/or growth equation.
- ⇒ To be confident and competent with the operations of a scientific calculator and its use.
- ⇒ To apply mathematical techniques to simple "real life" problems.

TOPICS OF STUDY

- ⇒ Linear Equations
- ⇒ Simple probability
- ⇒ Quadratic Equations, Remainder Theorem, Equation of a Circle and Inequalities
- ⇒ Indices, Exponential and Logarithmic Functions
- ⇒ Binomial Expansions, Sequences and Series
- ⇒ Trigonometric Functions
- ⇒ Calculus - differentiation
- ⇒ Calculus - integration
- ⇒ Further Differentiation
- ⇒ Functions
- ⇒ Further Trigonometry
- ⇒ Further Integration
- ⇒ Vectors
- ⇒ Differential Equations
- ⇒ Numerical Methods
- ⇒ The Normal Distribution

ASSESSMENT

	What is Assessed?	Topics A-H
	Duration/Word Count	2 hours 10 minutes
	Total Marks	100
	Rubric	Approximately 13 compulsory questions of varying length.
	Contribution to Overall Grade	30%

	What is Assessed?	Topics A-P
	Duration/Word Count	2 hours 40 minutes
	Total Marks	125
	Rubric	Approximately 15 compulsory questions of varying length.
	Contribution to Overall Grade	70%

ASSESSMENT OBJECTIVES

	Description	End of Semester 1 Test (30%)	Exam (70%)	Overall
AO1	Knowledge & Understanding: Knowledge of mathematical techniques.	50-60%	50-60%	≈55%
AO2	Application and Communication: Recognising the appropriate procedure and applying combinations of skills and techniques.	40-50%	40-50%	≈45%

GENERAL LEARNING OUTCOMES

On successful completion of this module, a student will be able to:

Knowledge and understanding	⇒ Recognise, recall and apply specific mathematical facts, principles and techniques. ⇒ Select, organise and present relevant information clearly and logically. ⇒ Select and apply appropriate mathematical techniques to solving problems.
Intellectual skills	⇒ Apply mathematical techniques to problems from a variety of relevant discipline areas. ⇒ Present and interpret data in tables, diagrams and graphs. ⇒ Carry out calculations using a formula booklet and calculator. ⇒ Give answers to an appropriate degree of accuracy (i.e. to a specified number of significant figures or decimal places). ⇒ Discuss and interpret results obtained, including an estimate of accuracy.
Practical skills	⇒ Specify what data are required for a given task. ⇒ Recognise what aspect of Mathematics is required in a practical Engineering situation.
Transferable skills	⇒ Study independently and make personal notes for problem-solving and revision purposes. ⇒ Source and retrieve information from a variety of locations such as textbooks, websites etc. ⇒ Select and employ problem-solving skills (description, formulation, solution/analysis, interpretation). ⇒ Manage and present data in a variety of formats.

TEACHING, LEARNING AND ASSESSMENT GUIDANCE

Teachers should use a range of different learning and delivery styles in order to give students experience of the types of approach they will encounter in an NCUK University e.g. lectures, tutorials, seminars, laboratory practicals etc. Appendix A (Exemplar Teaching Plan) suggests a delivery format designed to facilitate teaching and learning in each topic. This is provided as a guide and teachers are at liberty to diverge from this plan. The Study Centre may also choose to timetable more than the minimum number of teaching hours (4 hours per week) to meet the needs and abilities of students.

Teachers should take every opportunity to use case study and data response or practice material to develop the ability of students to apply their knowledge and develop their skills of handling and evaluating information. Tutorials should involve both group and individual activities, with a strong emphasis on applying knowledge from lectures and independent study to problem-solving. It is important for all students to have opportunities to speak in English during each tutorial. Suggested activities include students being encouraged to explain in English their answers in pairs or small groups, students providing answers to the whole class whilst standing at the front of the group, student-led presentations, and group activities that require discussion and debate.

Students will have different backgrounds in the subject and it will be necessary to provide opportunities for directed independent study, so as to allow each student to develop at their own pace to reach the required level for the assessments. Activities (homework) for independent study should be set weekly. It is advisable to also consult with EAP teachers when planning homework as there may be subject tasks which can support the development of academic English (and vice-versa).

DEVELOPING INDEPENDENT LEARNING

It is important that the learners are given the guidance and support to develop from pupils within a teacher-led learning environment, to students who feel confident taking ownership of their learning through independent study. As students may have an educational background where concepts of teaching and learning are different to those prevalent in NCUK Universities, aspects of independent learning should be introduced gradually throughout the programme.

Below are suggested examples of ideas which could be used to encourage independent learning:

Acknowledging Failure and Success

A fear of failure or making mistakes can be a significant obstacle which prevents learners from being independent. It is therefore important to acknowledge the benefits of failure with regard to learning early in the module delivery. This could be explored in more depth through practical examples and calculations in which failure leads to success. Teachers should ensure that students are also praised, particularly through formative feedback, and encourage learners to demonstrate development from their past mistakes.

Classroom Reversal

Where possible, teachers might entrust the teaching, practice and revision of sub-topics or items from the syllabus to students themselves. Sharing the responsibility of teaching with students can be a more motivating way to encourage them to take their first steps into independent study than being asked to study independently for purely personal reasons. It also provides a strong reason why independent learning

should be carried out (in this case, to support their peers) and a visible outcome of their independent work (the performance of their peers).

A demonstration of possible student-led tasks and activities can be a useful way to introduce this concept within the classroom. A clear schedule for student-led sessions can also help to make it routine. Sessions could be led by groups initially before asking students to lead them individually when they have grown in confidence.

PAST ASSESSMENTS

Many learners can find it reassuring to revisit past assessments to support their preparation for formative and summative assessments. A selection of past Coursework, End of Semester 1 Test and Final Examination papers are available for this purpose via the NCUK Teachers' Hub. It is important that students are guided to sections and questions which will support the meeting of current learning outcomes or that past assessments are appropriately adapted to ensure their fit to the current syllabus.

While classroom time can be allocated to the review of and use of past papers, students should also have the opportunity to make use of these for independent study. Teachers should work with colleagues to decide which papers may be kept aside for use in formative assessment or class time and which to release to students for independent study.

There are a number of ways past assessments can be used to support learning in addition to recalling knowledge. These include:

- ⇒ An exploration of incorrect or incomplete answers which could highlight gaps in understanding and help to inform future teaching and learning
- ⇒ Identifying whether the correct structure is being used to answer extended answers
- ⇒ Practicing answers to certain questions within particular time constraints
- ⇒ Distinguishing different command words, such as 'explain', 'describe' and 'evaluate'
- ⇒ Allowing students to notice weaknesses in their own knowledge or answers by allowing them to self-mark with the assessment criteria
- ⇒ Allowing students to establish a greater familiarity with the assessment criteria and what they are expected to do to achieve a certain grade

SUPPORTING ENGLISH LANGUAGE DEVELOPMENT

It is important for the development of academic literacies that the delivery of the subject material is integrated with the English for Academic Purposes (EAP) module (or EAP for Proficient Users or Research and Communication Skills modules where appropriate). Regular communication between the subject module teacher(s) and the EAP, EAPPU or RCS teacher(s) will provide a basis on which to support and guide students. Students will benefit from collaborative activities, where the subject module and EAP/EAPPU/RCS teachers jointly deliver classes in relation to activities such as essay or report writing style and using academic sources.

There are also a number of simple tasks and approaches which can be built into this module which can support language development. These include:

Introducing New Vocabulary

When students are introduced to new vocabulary in class, ensure that these words are not introduced in isolation, but in a chunk or phrase within a wider understood context. As an example, if students were given the following extract from the Economist (2021), they may be unfamiliar with the word 'ailments':

The cholera pandemic of the early 1830s hit France hard. It wiped out nearly 3% of Parisians in a month, and hospitals were overwhelmed by patients whose ailments doctors could not explain.

If this word was then to be explored and recorded, it would be better to look at this chunk:

...hospitals were overwhelmed by patients whose **ailments** doctors could not explain.

By giving students a clear context, it is easier to initially discern meaning while also making it more memorable for revision. By keeping the word within a larger chunk, there are also a number of grammatical and lexical relationships which can be established by learners:

- ⇒ Common collocations; explain ailments, a patient's ailments,
- ⇒ The inclusion of 'ailments' in the semantic field of medicine / health.
- ⇒ The determiner 'whose' implies that 'ailments' is a noun and the 's' that it is a plural, countable noun.
- ⇒ Wider examples of past passive and defining relative clauses.

Although it should not be expected that this level of exploration is conducted within a subject module, highlighting new vocabulary in chunks does give language learners a greater opportunity to notice features of language at their own speed. It also provides a grammatical template which students can build on when writing their own answers.

Highlighting Stressed Syllables

One way to support the development of students spoken fluency and overall rhythm is to draw attention the stressed syllable within key, subject-specific vocabulary. This can be done very simply on the board in a number of ways and you may wish to discuss this with the EAP/RCS teacher(s) at your Study Centre in order to align your preferred method(s). As you may be students' main model for how to pronounce subject-specific vocabulary, it is essential they have a clear record which they can refer to during independent study.

Encouraging Functional & Discursive Language

When completing written and spoken tasks, such as group discussions or the presentation of ideas within class, you may wish to encourage students to use functional or discursive language that will help them to structure and explain their ideas with greater coherence and clarity. For instance, this may include language for agreeing /disagreeing, clarifying what they have said when they are unclear, or how to highlight a particular point. This could be achieved by:

- ⇒ Asking students to take out, look at and use language explored for this purpose within their EAP classes. Collaboration with the EAP teacher(s) to decide when in the module particular language will be covered and for what purpose will greatly support students' language development.

- ⇒ Make functional and discursive language relevant to your subject available in the classroom for students to use. Adding useful language to walls or 'vocabulary mats' on desks can remind students of language available to them while also encouraging independence.

Allowing Learners Time to Speak

Although some students may be reluctant to speak in class, it is also important to remember that less proficient speakers require time to construct utterances and may need some time to prepare themselves. While silence in a classroom can sometimes feel awkward, try to avoid filling this silence with teacher talk time, particularly when you are expecting answers from students. You may want to prepare students to speak by giving them advance warning, for example, by giving the class time to make notes before being asked to speak. Alternatively, you may wish to select students to answer particular questions in advance.



FORMATIVE ASSESSMENT

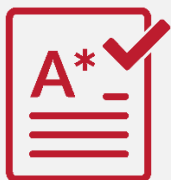
It is important that students are given the opportunity to engage in and submit formative assessments and receive feedback on this work. Formative assessment should be designed to inform students of their progress and enable them to develop and practice coursework and examination skills.

In order to assist teachers in setting formative assessment, NCUK provides the following for this module:

- ⇒ Exemplar assessments – a full set of exemplar assessments with mark schemes (End of Semester 1 Test, Coursework and Final Examination) are downloadable from the Teachers' Hub. These can be used as mock assessments for students throughout the academic year.
- ⇒ Past exam questions – a set of past examination questions is accessible for teachers via the Teachers' Hub. Teachers can generate their own papers for use as practice exams, pick out specific questions to work through with students in class, or set as homework.

In addition to the above, teachers should set regular homework tasks that allow students to understand their strengths and weaknesses and should provide advice to students on how to improve in areas of weakness.

Beyond developing students' skills, the crucial element of formative assessment is the feedback provided by the teacher. In some instances, it may also be appropriate for students to provide feedback on each other's work in a group setting.



SUMMATIVE ASSESSMENT

The breakdown of summative assessment for this module and the rubrics for each component are given on page 3 of this document. All summative assessments are provided by NCUK.

End of Semester 1 Test

The End of Semester 1 Test will cover topics from Semester 1 only and will take place under examination conditions. Teachers at the Study Centre will be responsible for marking the End of Semester 1 Tests in accordance with the NCUK issued mark scheme.

Examination

The examination will cover topics from both semesters. Teachers at the Study Centre will be responsible for marking the Examination in accordance with the NCUK issued mark scheme.

An NCUK approved calculator is permitted in the examination (refer to NCUK policy '*Calculator Regulations*').

It is the Study Centre's responsibility to provide the following materials for the End of Semester 1 test and Examination:

- ⇒ Graph Paper
- ⇒ Formula Booklet (refer to *Appendix C* of this syllabus)

Supporting Documents

It is essential that coursework and examinations are administered in accordance with NCUK regulations. Please refer to the following sections of the NCUK Academic Handbook for details:

- ⇒ Coursework Administration and Regulations
- ⇒ Centre Marking and Recording Results
- ⇒ Academic Misconduct Policy
- ⇒ Examination Administration

RESOURCES



Core Textbooks

CGP Books (2017). Mathematics for A level The Textbook

ISBN: 9781782947233 (Print)

Electronic version available to purchase [here](#)

CGP Books (2018), *AS and A level Further Maths for Edexcel Complete Revision and Practice*, Coordination Group Publications Ltd (CGP)

ISBN: 9781782948698 (Print)

Each copy comes with a free online edition.



Further Reading

An introduction to Mathematics written in the style of a travelogue:

Bellos, A. (2020), *Alex's Adventures in Numberland*. Bloomsbury Publishing

ISBN: 9781526623997 (Print)

ISBN: 9781408811146 (eBook)

Centre for Innovation in Mathematics Teaching A-level Course Material:

<https://www.cimt.org.uk/projects/mepres/alevel/alevel.htm>



Useful Websites

Free mathematics software:

<http://www.geogebra.org/>

<https://www.desmos.com/>

A variety of resources sorted by topic:

<https://www.mathcentre.ac.uk/>

A level maths resources including links to Engineering:

<https://www.stem.org.uk/resources/curated-collections/secondary-and-level-mathematics-0>

A level maths revision resources:

<https://app.senecalearning.com/courses?Price=Free&Subject=Maths&Age+Group=A+Level>

For students and teachers to pose or answer any maths problem at any level:

<http://math.stackexchange.com>



Teacher Resources

NCUK Teacher's Hub – accessible via the NCUK learning and assessment platform – please contact your Study Centre Academic Manager for access.

MODULE CONTENT

All topics must be covered. Appendix A provides an Exemplar Teaching Plan which indicates the proportion of time to be spent on each topic and activities to support student learning.

The table below details the topics and sub-topics to be covered. A set of specific learning outcomes (SLOs) are given for each topic.

TOPIC A – LINEAR EQUATIONS		
CONTENT	SLOs	
⇒ The equation of a line, parallel and perpendicular lines. ⇒ Solving pairs of simultaneous equations using elimination, substitution and graphical methods.	A1	Find the equation of a straight-line using co-ordinate geometry.
	A2	Find parallel and perpendicular lines and sketch appropriate graphs.
	A3	Find the distance between two points and the co-ordinates of the midpoint.
	A4	Solve pairs of simultaneous equations using elimination, substitution and graphical methods.

TOPIC B – SIMPLE PROBABILITY		
CONTENT	SLOs	
⇒ Define probability, use sample space diagrams to help calculate probabilities. ⇒ Combining probabilities and using tree diagrams. ⇒ (Knowledge of conditional probability such as $P(A B)$ is not expected in this module.) ⇒ (The use of the symbols $\cap$ and $\cup$ will not be expected in this module).	B1	Find the probability of a single event.
	B2	Recognise that $P(A)$ and $P(A')$ mean the probabilities of event A occurring and event A not occurring respectively.
	B3	Find, for two events A and B , the probabilities of both A and B occurring, and the probabilities of either A or B occurring.
	B4	Construct and use a tree diagram.

TOPIC C – QUADRATIC EQUATIONS, REMAINDER THEOREM, EQUATION OF A CIRCLE AND INEQUALITIES		
CONTENT	SLOs	
⇒ Quadratic Functions: factorising, completing the square and using the quadratic formula. ⇒ Remainder Theorem: simple algebraic division; use of the Factor Theorem and the Remainder Theorem. Graphs of quadratic and cubic functions. Geometrical interpretation of algebraic solutions of equations. ⇒ Equation of a Circle: in the form $(x - a)^2 + (y - b)^2 = r^2$. ⇒ Inequalities: manipulating inequalities, solving linear and quadratic equations and inequalities.	C1	Solve quadratic equations using factorising, completing the square and the quadratic formula.
	C2	Carry out the process of completing the square to locate vertices (turning points) of graphs.
	C3	Use the discriminant to determine the number of real roots.
	C4	Use surds to give exact solutions.
	C5	Use algebraic division by a monomial or quadratic function.
	C6	Sketch the graphs of quadratic and cubic functions.
	C7	Use the Factor Theorem. Use the Remainder Theorem to determine the remainder when a polynomial is divided by $(ax + b)$.
	C8	Find the radius and co-ordinates of the centre of a circle given its equation and vice versa.
	C9	Recognise that the radius of a circle is perpendicular to a tangent.
	C10	Solve by substitution a linear and quadratic pair of simultaneous equations and apply this to finding the intersection of a line and a curve/circle.
	C11	Recognise and solve linear/quadratic equalities and inequalities.
	C12	Use algebraic and graphical methods to solve inequalities.
	C13	Recognise and distinguish between open and closed intervals (candidates will be expected to understand the word “integer”).

TOPIC D – INDICES, EXPONENTIAL AND LOGARITHMIC FUNCTIONS		
CONTENT	SLOs	
⇒ Laws of Indices for all rational exponents. ⇒ Exponential function: ○ exponential function and its graph ○ introduction to rates of growth ○ solution of equations involving exponential functions ⇒ Logarithmic function: ○ rules and manipulation of logarithms ○ logarithmic function and its graph ○ relationship between exponential / logarithmic functions ○ solution of equations involving either exponential or logarithmic functions	D1	Use and apply the equivalences $x^a \times x^b \equiv x^{a+b}$, $x^a \div x^b \equiv x^{a-b}$, $(x^a)^b \equiv x^{ab}$ $x^{-n} \equiv \frac{1}{x^n}$, $x^{a/b} \equiv \sqrt[b]{x^a}$
	D2	Express solutions in exact form.
	D3	Use a calculator to evaluate exponential and logarithmic expressions.
	D4	Sketch the graphs of $y = e^x$, $y = e^{-x}$ and $y = a^x$.
	D5	Apply exponential functions to problems.
	D6	Apply the rules of logarithms to problems.
	D7	Change the base of a logarithm.
	D8	Solve equations involving exponential and logarithmic functions including those involving “hidden” quadratics.

TOPIC E – BINOMIAL EXPANSIONS, SEQUENCES AND SERIES		
CONTENT	SLOs	
⇒ Binomial expansions: ○ Pascal's triangle ○ factorials ○ binomial expansion (positive integer powers, binomial coefficient notation, evaluation of specific terms) (Candidates could be asked to write down the first few terms in ascending powers of x of an expansion with $n \gg 5$. The meaning of "ascending" or "descending" will always be given. They will be expected to understand the word "coefficient") ⇒ Sequences and series: ○ sequences ○ series ○ sigma notation ⇒ Finite Arithmetic Progressions (AP) and series including sum. ⇒ Geometric Progressions (GP) and series including sum. ⇒ Convergence and divergence of geometric series.	E1	Expand $(1 + x)^n$ for small positive integer n .
	E2	Use Pascal's triangle to find binomial coefficients.
	E3	Expand $(a + b)^n$ where n is a small positive integer.
	E4	Evaluate specific terms of the expansion of $(a + b)^n$ where n is a positive integer.
	E5	Understand the idea of a sequence of terms using general formulae and recurrence relations.
	E6	Use sigma notation for series representations.
	E7	Recognise and sum a finite arithmetic series.
	E8	Recognise and sum a geometric series.
	E9	Define, explain and test for convergence of a series.
	E10	Use an AP or GP to solve practical problems.

TOPIC F – TRIGONOMETRIC FUNCTIONS		
CONTENT	SLOs	
⇒ Angles (degree/radian measure). ⇒ Trigonometric ratios, trigonometric functions (sine, cosine, tangent) and their graphs. ⇒ Solutions of simple trigonometric equations. ⇒ The identity $\cos^2 x + \sin^2 x \equiv 1$. ⇒ $\tan x = \frac{\sin x}{\cos x}$.	F1	Convert from radians to degrees and vice versa.
	F2	Recognise exact values of the trigonometric functions of 0, 30, 45, 60 and 90 degrees and their radian equivalents.
	F3	Find sin, cos and tan of any angle and sketch their graphs.
	F4	Use the area of a triangle formula $\frac{1}{2}ab \sin C$.
	F5	Calculate inverse trigonometric functions.
	F6	Find particular solutions of simple trigonometric equations. (These equations will take the form: $a \sin bx = c$; $a \cos (x + b) = c$; $a \tan^2 bx = c$ over any range.)
	F7	Recognise that $\cos^2 x + \sin^2 x \equiv 1$ and $\tan x = \frac{\sin x}{\cos x}$ (extensive use of these properties is not required at this stage).
	F8	Apply the sine and cosine rules to an arbitrary triangle.

TOPIC G – CALCULUS - DIFFERENTIATION		
CONTENT	SLOs	
⇒ Gradients of tangents and normals to curves, limit form, polynomial rules (inc. First Principles). ⇒ Derivatives of simple functions (exponential, log, trigonometric. The trigonometric functions are $\sin x$, $\cos x$ and $\tan x$ only). ⇒ Use of Formula Booklet (see Appendix C). ⇒ Using derivatives to help sketch curves. ⇒ Equations of tangents and normals. Maxima, minima and points of inflexion which are stationary points. ⇒ Use of the second derivative.	G1	Evaluate the gradient of a curve at a point.
	G2	Recognise and explain the notation $\frac{dy}{dx}$ and $f'(x)$.
	G3	Sketch the derivative graph $\frac{dy}{dx}$.
	G4	Apply the limit formula to simple functions (first principles). (This will be confined to single powers of x .)
	G5	Differentiate any polynomial and any power of x .
	G6	Use the formula booklet to obtain derivatives of standard functions, including e^{kx} where k is a constant.
	G7	Explain second-derivative notation.
	G8	Apply second derivatives to practical problems.
	G9	Find stationary points for a given function.
	G10	Distinguish between local maximum, local minimum and point of inflexion which are stationary points including application to curve sketching. (Points of inflexion will be identified only in instances of zero gradient.)
	G11	Apply SLOs G8 and G9 to practical optimisation problems.
	G12	Obtain the equation of tangent and normal of a curve at a specified point.

TOPIC H – CALCULUS - INTEGRATION		
CONTENT	SLOs	
⇒ Inverse of differentiation, standard integrals (polynomial, trigonometric, exponential), indefinite and definite integration. (The trigonometric functions are $\sin x$ and $\cos x$ only.) ⇒ Area under a curve.	H1	Identify integration as the inverse of differentiation.
	H2	Use formula booklet to determine indefinite integrals including e^{ax} and $\frac{1}{ax+b}$ where a and b are constants.
	H3	Form and explain the definite integral.
	H4	Evaluate definite integrals.
	H5	Find the area between a curve and the x -axis, including areas partly above and partly below the axis.
	H6	Find the area between a curve and a straight line and between two curves.

TOPIC I – FURTHER DIFFERENTIATION		
CONTENT	SLOs	
⇒ Product, quotient and chain rules (composite functions). ⇒ Implicit differentiation.	I1	Apply the product rule, quotient rule and chain rule.
	I2	Find $\frac{dy}{dx}$ of an implicit function and apply to inverse trigonometric functions e.g. $y = \sin^{-1}(ax + b)$.
	I3	Recognise the result $\frac{d}{dx} a^x = a^x \ln a$.

TOPIC J – FUNCTIONS		
CONTENT	SLOs	
⇒ Language: definition of function/terms: many-one, one-one, one-many, many-many, mapping, domain, range, odd, even, periodic. ⇒ Transformations: effect of transformations on graphs. ⇒ Sketch graphs of transformed functions. ⇒ Compositions: composition of two functions, conditions for inverse to exist. ⇒ Modulus function: Investigate relationship between graphs of $y = f(x)$ and $y = f(x)$.	J1	Explain the terms many-one, one-one, one-many, many-many.
	J2	Describe the terms mapping, domain, range, odd, even, periodic.
	J3	Implement simple transformations (stretch parallel to x – and y – axes, translation by a vector, reflection in x – and y – axes).
	J4	Compose two functions.
	J5	Determine the inverse of a function.
	J6	Sketch graphs of the modulus of linear functions.
	J7	Solve linear modulus equations.

TOPIC K – FURTHER TRIGONOMETRY		
CONTENT	SLOs	
⇒ Simple trigonometric identities (double angle, sums, differences) and their use in calculus. ⇒ Inverse trigonometric functions. ⇒ Use of a calculator for trigonometric and inverse trigonometric functions, solutions of harder trigonometric equations.	K1	Recognise and use the definitions of secant, cosecant and cotangent and their graphs.
	K2	Work with inverse trigonometric functions, their graphs and restricted domains.
	K3	Use and apply the addition formulae and double angle formulae.
	K4	Use and apply the identities $\sec^2\theta \equiv 1 + \tan^2\theta$ and $\csc^2\theta \equiv 1 + \cot^2\theta$.
	K5	Use the formulae given in SLOs F7, K3 and K4 to prove trigonometric identities.
	K4	Use the formulae given in SLOs F7, K3 and K4 to solve trigonometric equations over a given interval.
	K5	Use the formulae given in SLOs F7, K3 and K4 in calculus.
	K6	Use trigonometric identities to prove/show various results.

TOPIC L – FURTHER INTEGRATION		
CONTENT	SLOs	
⇒ Integration by substitution: Change of variables (inc. trigonometric substitutions and identities). ⇒ Use of Formula Booklet (see Appendix C). ⇒ Partial fractions: linear factors, repeated linear factors, quadratic factors, improper fractions and integration by partial fractions. ⇒ Integration by parts (examination questions will involve no more than two applications). ⇒ Evaluation of volume of revolution.	L1	Integrate standard functions which may be polynomials, trigonometric functions or exponential functions.
	L2	Recognise the integral of $\frac{f'(x)}{f(x)} = \ln(f(x)) + c$.
	L3	Clarify what is meant by the term “partial fraction”.
	L4	Find partial fractions for linear, repeated linear and quadratic factors in form $ax^2 \pm b$.
	L5	Find partial fractions for improper fractions.
	L6	Carry out integration using partial fractions.
	L7	Use substitution (including trigonometric forms) to evaluate indefinite and definite integrals (other than in simple cases, the substitution will normally be given).
	L8	Carry out integration by parts.
	L9	Evaluate the volume of revolution of simple solids (formed by rotation about the x – axis only).

TOPIC M – VECTORS		
CONTENT	SLOs	
⇒ Representation and algebra: 2D and 3D vectors. ⇒ Addition, subtraction, multiplication by a scalar, unit vectors. ⇒ Cartesian components, position vectors. ⇒ Scalar product: Definition, calculation. ⇒ Vector equation of a line, angle between two lines, intersection of two lines.	M1	Define and demonstrate the terms <i>scalar</i> and <i>vector</i> .
	M2	Apply vector addition and subtraction.
	M3	Write 2D vectors in <i>i, j</i> notation and use a calculator to find the resultant vector.
	M4	Write 3D vectors in <i>i, j, k</i> notation and find the magnitude of a 3D vector.
	M5	Evaluate the unit vector.
	M6	Clarify the difference between a general vector and a position vector.
	M7	Find scalar products and examine Engineering applications of a scalar product.
	M8	Develop and evaluate the vector equation of a line.
	M9	Calculate the angle between two lines.
	M10	Find the point of intersection of two lines.
	M11	Calculate the perpendicular distance from a point to a line.

TOPIC N – DIFFERENTIAL EQUATIONS		
CONTENT	SLOs	
⇒ Differentiation Techniques: the use of $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$. ⇒ Formation of simple differential equations (may include related rates of change). ⇒ Solution of first order differential equations with separable variables.	N1	Use and apply $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.
	N2	Find the order of a differential equation.
	N3	Develop simple first-order models of problems.
	N4	Solve first-order differential equations by direct integration.
	N5	Solve first-order differential equations by separation of variables.
	N6	Solve Engineering applications with initial conditions.
	N7	Solve differential equations with an exponential rate of change.

TOPIC O – NUMERICAL METHODS		
CONTENT	SLOs	
⇒ Location of a root by change of sign. ⇒ Using the Newton-Raphson method. ⇒ Numerical methods to find an approximate value of a definite integral. (Functions will be continuous.)	O1	Locate the roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval of x in which $f(x)$ is continuous.
	O2	Solve equations using the Newton-Raphson method.
	O3	Use the trapezium rule to find an approximate value of a definite integral.
	O4	Use Simpson's Rule to find an approximate value of a definite integral.
	O5	Deduce when the trapezium rule gives an overestimate or an underestimate.
	O6	Explain how the accuracy of the trapezium rule and Simpson's Rule can be improved.
	O7	Find the percentage error in the approximation of a definite integral.

TOPIC P – THE NORMAL DISTRIBUTION		
CONTENT	SLOs	
⇒ The Normal Distribution and its applications. ⇒ Confidence Intervals.	P1	Use the Normal distribution in an Engineering context. (Candidates will be expected to do this on a calculator.)
	P2	Set up a confidence interval for a mean where the background distribution of any samples will be Normal with a known standard deviation.

APPENDIX A - EXEMPLAR TEACHING PLAN

Please note that the table below allocates a minimum of 4 hours contact time per week. Study Centres may need to schedule additional time depending on the ability of each cohort.

Method of delivery notation:

L – Lectures

E – Exercises

T – Tutorials

O – Other (specified)

Week	Hours	Topic	Teacher Guidance/Suggested Activities
1	1	A - Linear Equations	Identify the gradient and intercept in the equation of a straight line; calculate gradients of the normal. Find the distance between two points and the co-ordinates of the midpoint. (L/E) There may be considerable variation in the algebraic ability of students. Any practice in this area will probably be good for independent and directed study.
	1	A - Linear Equations	Find equations of other lines which are parallel to or perpendicular to the equation of a particular line. Solve pairs of linear simultaneous equations using elimination, substitution and graphical methods. (L/E)
	1	A - Linear Equations	Continue with simultaneous equations. Carry out practice examples on linear equations. (Mostly E)
	1	B - Simple Probability	Evaluate the probability of a single event. Familiarity with the notation $P(A)$ and $P(A')$ and realise that $P(A') = 1 - P(A)$. (L/E/T)
2	1	B - Simple Probability	For two events A and B , work out the probability of both happening, and of either happening. (Set notation is not required at this level – neither will candidates be expected to be aware of mutual exclusivity). Construct a tree diagram and use it to work out combined probabilities. (Mostly L)
	1	C - Quadratic Functions and Equations	Carry out a completing the square process and be able to sketch the graph of a quadratic function. (L/E) Handling algebra is often a weak spot, so what was said about independent and directed study in the previous week probably applies here.

	1	C - Quadratic Functions and Equations	Solve quadratic equations by factorising, completing the square and using the quadratic formula. Candidates must be able to present answers in surd form and understand the significance of the discriminant (but will not be expected to evaluate the size of coefficients in the original equation which give, for example, two real roots). (L/E) If time runs short, there is scope for continuing the process of solving by factorisation in the following week.
	1	C - Quadratic Functions and Equations	Continuation of previous session. (T & mostly E)
3	1	C - Further simultaneous equations	Solve by substitution two equations – one of which is quadratic and the other linear. (E & mostly L)
	1	C - Equation of a Circle	The form $(x - a)^2 + (y - b)^2 = r^2$. Find the radius and coordinates of the centre of a circle given its equation and vice versa. Know that the radius of a circle is perpendicular to a tangent. Link with previous lesson on solving equations e.g. find the points of intersection between a line and a circle.
	1	C - Inequalities	Solve linear and quadratic inequalities, either by using algebra or graphical methods. Recognise open and closed intervals and understand what an integer is. (L/E)
	1	C - Remainder Theorem	Use the Factor Theorem to show a given monomial is a factor of a polynomial and the Remainder Theorem to find its remainder upon division if it is not a factor. Candidates will be expected to divide a polynomial by a monomial or quadratic expression, and to be able to factorise an expression completely. (L/E)
4	1	C - Remainder Theorem	Continuation of previous session. (Mostly E)
	1	D - Indices	Know that $x^a \times x^b \equiv x^{a+b}$, $x^a \div x^b \equiv x^{a-b}$, $(x^a)^b = x^{ab}$. $x^{-n} \equiv \frac{1}{x^n}$ and $x^{a/b} \equiv \sqrt[b]{x^a}$. (L/E)
	1	D - Indices	Find the exact solutions of equations which use indices. (L & mostly E)
	1	D - Exponential Functions	Understand what an exponential is and be able to sketch the graphs of $y = e^x$, $y = e^{-x}$ and $y = a^x$. (Mostly L) Investigating exponential growth and decay could provide good material for independent or directed study.
5	1	D - Exponential Functions	Relate exponential change to a real situation e.g. population growth and decomposition of a solid into a liquid or a gas. (L/E/T)

	1	D - Logarithmic Functions	Understand how a logarithm behaves and compare it to real-life situations such as the Richter and pH scales. (Mostly L)
	1	D - Logarithmic Functions	Establish the connection between logarithms and exponentials. (L/E) Practice at using and manipulating logs would be invaluable time spent for independent and directed study.
	1	D - Logarithmic Functions	Use the logarithmic laws and apply them to solve equations and simplify expressions. (L & mostly E)
6	1	D - Logarithmic Functions	Consolidate on previous work and practice in the use of logarithms. (E/T)
	1	E - Binomial Expansions	Revise expansion of quadratic functions. Use Pascal's Triangle to find binomial coefficients and be able to expand completely $(1+x)^n$ and $(a+b)^n$ where n is a small positive integer (normally not more than 5). (L/E)
	1	E - Binomial Expansions	Continuation of previous session. (Mostly E)
	1	E - Binomial Expansions	Consolidate the previous two sessions. Candidates will be expected to pick out single terms in the expansion of $(a+b)^n$ where n is any integer. Candidates must also know what a coefficient is. (L/E/T)
7	1	E - Progressions	Start on progressions and introduce the difference between an Arithmetic Progression (AP) and a Geometric Progression (GP). Identify the first term and common difference in an AP. (E & mostly L)
	1	E - Progressions	Use the relevant formulae to find the n th term and the sum of the first n terms of an AP and a GP. (L/E) Candidates could be asked to find the first term of a GP, or how many terms are needed for geometric series, to exceed a certain value. This should only be done once logs have been covered.
	1	E - Progressions	Candidates should be able to find the common difference, common ratio and first term having been given the n th term or sum of the first n terms. (L & mostly E)
	1	E - Progressions	Use sigma notation and be able to generate a progression from a sigma expression. Generate the first few terms of a simple recurrence relation. Understand the idea of convergence and the conditions needed. Find the sum to infinity of a convergent series. (L/E)

8	1	F - Trigonometric Functions	Understand the definition of a radian; convert angles from degrees to radians and vice versa. Calculate the sin, cos and tan of any angle and be familiar with their graphs. (L/E)
	1	F - Trigonometric Functions	Know how to find the inverse of a trigonometric expression and solve simple trigonometric equations. Candidates will also need to be able to quote and recognise the exact values of the trigonometric functions of 0, 30, 45, 60 and 90 degrees and their radian equivalents. (L/E)
	1	F - Trigonometric Functions	Consolidate the previous two sessions and practice at examples – particularly solving the equations and identifying all the angles in each interval. (E/T) It is probably a case of the more time that students spend at practising examples in independent and directed study, the better.
	1	F - Trigonometric Functions	Apply the sine and cosine formulae to a non-right-angled triangle and use the formula for the area of a triangle $\frac{1}{2}ab \sin C$. Candidates should be familiar with the identity $\cos^2 x + \sin^2 x \equiv 1$ and with $\tan x = \frac{\sin x}{\cos x}$. (L/T)
9	1	G - Differentiation	Revision of finding the gradient of a straight line; understand the definition of the gradient of a curve at a point and, using first principles, find an expression for this gradient (using single integral powers of x only). (Mostly L)
	1	G - Differentiation	Using first principles, find gradient functions of other small powers of x . Introduce the idea of differentiating any power of x . (Mostly L)
	1	G - Differentiation	Use $\frac{dy}{dx}$ and $f'(x)$ and be able to differentiate any polynomial and any power of x . (L/E)
	1	G - Differentiation	Practice of differentiating polynomials and substituting in values of x . (Mostly E)
10	1	G - Differentiation	Extend differentiation to find the equations of a tangent and a normal to a curve at a specified point. (L/E)
	1	G - Differentiation	Extend the process to differentiation of exponentials (expressions of the form $y = e^{kx}$) and logarithms. (L/E)
	1	G - Differentiation	Differentiation of trigonometric expressions ($\sin x$, $\cos x$ and $\tan x$ only). Use of the formula booklet to obtain derivatives of standard functions. (L/E)
	1	G - Differentiation	Consolidate on the previous two weeks' work with plenty of practice at differentiation. (E/T)

			Plenty of scope here for practice in independent and directed study.
11	1	G - Differentiation	Find the second derivative of a function and be familiar with the notation $\frac{d^2y}{dx^2}$ and $f''(x)$. (L/E)
	1	G - Differentiation	Find the stationary points of a function. (L/E) Points of inflexion will be identified only in instances of zero gradient.
	1	G - Differentiation	Determine the nature of any stationary point. (L/E)
	1	G - Differentiation	Sketch the graph of a function once the turning points are known. Sketch the graph of a derivative. (Mostly E)
12	1	G - Differentiation	Extend the process of finding stationary values to practical optimisation problems. Typical cases could be to find the maximum volume of a solid which has a fixed surface area, and to find the minimum surface area of a solid with a fixed volume. (Mostly L) Other examples could be maximising an area of a shape with fixed perimeter.
	1	G - Differentiation	Extension of previous session. (L/E)
	1	G - Differentiation	More practice at optimisation problems and start of a general consolidation of differentiation. (Mostly E) In view of the size of this topic, a long consolidation will probably be necessary.
	1	G - Differentiation	Completion of the consolidation started the previous session. (E/T)
13	1	H - Integration	Identify integration as the inverse of differentiation. Integrate any power of x . (Mostly L)
	1	H - Integration	Integrate e^{kx} where k is a constant; integrate $\sin x$ and $\cos x$. (L/E)
	1	H - Integration	Realise the integration of $\frac{1}{x} = \ln x + c$. Use the formula booklet to determine indefinite integrals including integrals of the form $\frac{1}{ax+b}$ where a and b are constants. (L/E) Practice at integration would probably be time usefully spent in independent and directed study.
	1	H - Integration	Find the indefinite integral of any of the functions above. (Mostly E)

14	1	H - Integration	Evaluate definite integrals. (L/E)
	1	H - Integration	Apply definite integrals to finding the area between the curve and the x – axis. (L/E)
	1	H - Integration	Find more difficult areas, including those below the x –axis. (L & mostly E) Students should realise that areas below the x – axis will give negative values and appreciate the meaning of an integral giving zero.
	1	H - Integration	Extend integration to finding an area which is bound by two or more curves. (E/T)
15	End of Semester 1 Test		
16	1	I - Further Differentiation	Differentiate using the product rule. (L/E) e.g. differentiate $4x^3 \cos x$.
	1	I - Further Differentiation	Differentiate using the quotient rule. (L/E) e.g. differentiate $\frac{1-x^2}{1+x^3}$.
	1	I - Further Differentiation	Differentiate using the chain rule. (L/E) Examples of the chain rule are: Differentiate $(2 - x^3)^5$ and $e^{\sin x}$.
	1	I - Further Differentiation	Consolidate on the three previous sessions. Know the result $\frac{d}{dx} a^x = a^x \ln a$. (Mostly E) Practice at differentiation in independent and directed study time will be invaluable.
17	1	I - Further Differentiation	Understand what an implicit function is and how to apply differentiation. (Mostly L) In this module, candidates will be expected to draw on concepts learnt in topic G.
	1	I - Further Differentiation	Find $\frac{dy}{dx}$ of an implicit function which could involve use of the product rule. (L/E) Again, time spent practising in independent and directed study will be invaluable.
	1	I - Further Differentiation	Find the equations of a tangent and a normal at a point. (L/E)

	1	I - Further Differentiation	Find stationary values. (L/E)
18	1	J - Functions	Understand what a function is and learn the terminology detailed in specific learning outcomes J1 and J2. (Mostly L)
	1	J - Functions	Find the inverse of a function and explain why an inverse may not exist. Compose two functions. (L/E) Realise that the domain and range of the inverse of a function is just their interchanging in the original function.
	1	J - Functions	Carry out transformations on functions (stretch parallel to x – and y – axes, translation and reflection in x – and y – axes) and sketch the graphs. (L/E)
	1	J - Functions	Understand what a modulus is and sketch the graph of a modulus function. Solve a linear modulus equation, either by graph or by algebra. (L/E) This topic will be confined to linear functions.
19	1	K - Further Trigonometry	Introduce sec, cosec and cot and their graphs. Graphs of inverse trig functions. The bulk of the time will probably be spent practising examples. Independent and directed study time can be put to good use in this area.
	1	K - Further Trigonometry	Revise $\cos^2 x + \sin^2 x \equiv 1$ and $\tan x = \frac{\sin x}{\cos x}$ introduce $\sec^2 \theta = 1 + \tan^2 \theta$, $\csc^2 \theta = 1 + \cot^2 \theta$ (L/E)
	1	K - Further Trigonometry	The addition formulae $[\sin(A \pm B); \cos(A \pm B); \tan(A \pm B)]$ and double angle formulae $\sin 2A; \cos 2A; \tan 2A$. Apply these formulae to show simple identities. (L/E)
	1	K - Further Trigonometry	Solve equations in each interval. (L/T)
20	1	K - Further Trigonometry	Use the formulae to differentiate trigonometric functions. (Mostly E) Candidates could for example be asked questions along the lines 'show that the differentiation of $\tan x$ is $\sec^2 x$.' or 'differentiate $\frac{2\tan x}{1 - \tan^2 x}$ '
	1	K - Further Trigonometry	Use the formulae to integrate functions. (Mostly E)

			e.g. integrate $\sin^2 x$. As for the previous week, it is anticipated that much of the time will be spent doing practice. There is also good scope for independent and directed study.
	1	K - Further Trigonometry	Use trigonometric identities to prove/show various results. (E/T)
	1	K - Further Trigonometry	Consolidate on previous three sessions. (E/T)
21	1	L - Further Integration	Integrate standard functions. Recognise the integral of $\frac{f'(x)}{f(x)}$. (L/E) e.g. $\cos 5x$, e^{ax} , $\sec^2 3x$, $\frac{1}{ax}$.
	1	L - Further Integration	Continuation of previous session. (Mostly E)
	1	L - Further Integration	Integrate by change of variable. Integrals involving a substitution can be indefinite or definite. (L/E) Other than in simple cases, the substitution to be used will normally be given.
	1	L - Further Integration	Continuation of previous session. (Mostly E) Independent and directed study time can be put to good use in this area.
22	1	L - Further Integration	Resolve an expression into partial fractions. (Mostly L) The number of partial fractions will not exceed three.
	1	L - Further Integration	Resolve an improper fraction into partial fractions. (L/E) Denominators will be linear, repeated linear or quadratic [in the last case, the expression will be of the form $(ax^2 \pm b)$].
	1	L - Further Integration	Consolidate on previous two sessions. (E/T)
	1	L - Further Integration	Integrate an expression in partial fraction form. (L/E)
23	1	L - Further Integration	Integrate by parts. (L/E) The process will be applied not more than twice.
	1	L - Further Integration	Continuation of previous week. (Mostly E)
	1	L - Further Integration	Find the volume of revolution of simple solids. (L/E) Rotations will be about the x – axis only.

	1	L - Further Integration	General consolidation where candidates develop the skill of recognising what integration method is needed. (E/T) Practising examples in directed and independent study will probably be time well spent.
24	1	M - Vectors	Understand the difference between scalar and vector. Apply addition and subtraction to 2D and 3D vectors. (L/E)
	1	M - Vectors	Find the resultant of 2D vectors (and apply to Engineering situations). (L/E) The sine and cosine formulae may need to be applied.
	1	M - Vectors	Find the magnitude of 3D vectors. Evaluate, and understand what is meant by a unit vector. (L/E)
	1	M - Vectors	Understand the difference between a general vector and a position vector; develop the vector equation of a line. (L/E)
25	1	M - Vectors	Use the scalar product to find the angle between two lines. (L/E)
	1	M - Vectors	Find the point of intersection of two lines and provide evidence to show when two lines do not meet. (L/E) Practice at finding the point of intersection of two lines will be useful in directed and independent study time.
	1	M - Vectors	Calculate the distance between two points, and the shortest distance from a point to a line. (L/E)
	1	M - Vectors	Consolidate on previous three sessions. (L/E/T)
26	1	N - Differential Equations	Identify the order of a differential equation. Set up a differential equation in an engineering context if appropriate. (L/E) Candidates could be asked to solve a differential equation which involves an exponential rate of change.
	1	N - Differential Equations	Solve a differential equation by direct integration or separation of variables. Understand what is meant by a general solution. (L/E)
	1	N - Differential Equations	Continuation of previous session. (Mostly E)
	1	N - Differential Equations	Apply any boundary conditions. Solve differential equations with boundary conditions (in an engineering context where appropriate). (E/T)
27	1	O - Numerical Methods	Locate the roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval of x in which $f(x)$ is continuous.

			Apply the Newton-Raphson method to find a more accurate solution to an equation. (L/E) Candidates will not be required to confirm the continuity of a function. When using the trapezium rule and Simpson's Rule, candidates could be asked how accuracy can be improved (normally by using more ordinates) and appreciate when answers may be overestimates or underestimates when using the trapezium rule.
	1	O - Numerical Methods	Use the trapezium rule to estimate the value of a definite integral. (L/E)
	1	O - Numerical Methods	Use Simpson's Rule to estimate the value of a definite integral. (L/E)
	1	O - Numerical Methods	Evaluate definite integrals where possible and compare answers with approximations obtained using the trapezium rule or Simpson's Rule. (L/E/T) Candidates could be asked to find the exact value of an integral and then find the percentage error in their estimate.
28	1	P - The Normal Distribution	Introduce the Normal distribution and the relevant use of a calculator. (Mostly L) Students may be able to relate to an Engineering context where appropriate (possibly the diameter of a piston ring which has to be within certain limits so as to be classified as not too small and not too large and working out the percentage of rings falling into this category).
	1	P - The Normal Distribution	Convert data into standardised form. Be aware of the limitations of the Normal distribution (e.g. the assumption that there is no restriction on the values taken by x but this is not usually the case in reality). (L/E) Candidates should be aware of the Central Limit Theorem but will not be required to apply it.
	1	P - The Normal Distribution	Set up a confidence interval (any samples will be taken from a background distribution which is assumed to be Normal with a known standard deviation). Explain what is meant by a confidence interval. (L/E)
	1	P - The Normal Distribution	Consolidate on the work covered over the last three sessions. (E/T)
29	4	Revision	
30	Exam Week		

APPENDIX B – NOTATION LIST

The notation below may be used in IFY Maths Engineering assessments without explanation. If forms of notation other than those listed are used in an assessment they will be defined within the question.

∞	Infinity
a	First term for an arithmetic or geometric sequence
d	Common difference for an arithmetic sequence
r	Common ratio for a geometric sequence
S_n	Sum to n terms of a sequence
S_∞	Sum to infinity of a sequence
$\sum_{i=1}^n a_n$	$a_1 + a_2 + \dots + a_n$
$\sqrt{a}$	the non-negative square root of a
$ a $	the modulus of a
$\frac{dy}{dx}$	the derivative of y with respect to x
$f'(x)$	the derivative of $f(x)$ with respect to x
$\frac{d^2y}{dx^2}$	the second derivative of y with respect to x
$f''(x)$	the second derivative of fx with respect to x
$\int y \, dx$	the indefinite integral of y with respect to x
$\int_a^b y \, dx$	the definite integral of y with respect to x between the limits $x=a$ and $x=b$
$\binom{n}{r}, {}^nC_r$	the binomial coefficient
$f^{-1}(x)$	the inverse function of the function fx
$f[g(x)]$	the composite function of g and f
$\sin, \cos, \tan$ $\csc, \sec, \cot$	the trigonometric functions
$\sin^{-1}, \cos^{-1}, \tan^{-1}$	the inverse trigonometric functions
$P(A)$	probability of the event A
A'	complement of the event A
z	standardized normal random variable
i, j, k	unit vectors in the directions of the cartesian coordinate axes

APPENDIX C - FORMULA BOOKLET

Refer to separate Formula Booklet.

It is the Study Centre's responsibility to print "clean" (new) copies of the Formula Booklet for the End of Semester 1 Test and Final Examination.

Syllabus Update History: NCUK International Foundation Year Mathematics (Engineering)

Release Date	Version No.	Summary of Main Changes	Author
June 2023	4.3	Updated for 2023/24	Product Team
May 2022	4.2	Updated for 2022/23	Product Team
Sept 2021	4.1	Link to electronic version of textbook updated after change by publisher	Product Team
June 2021	4.0	Full review	Debby Gill, Peter Davies & Product Team

This syllabus is valid for the 2023-24 academic year only.

